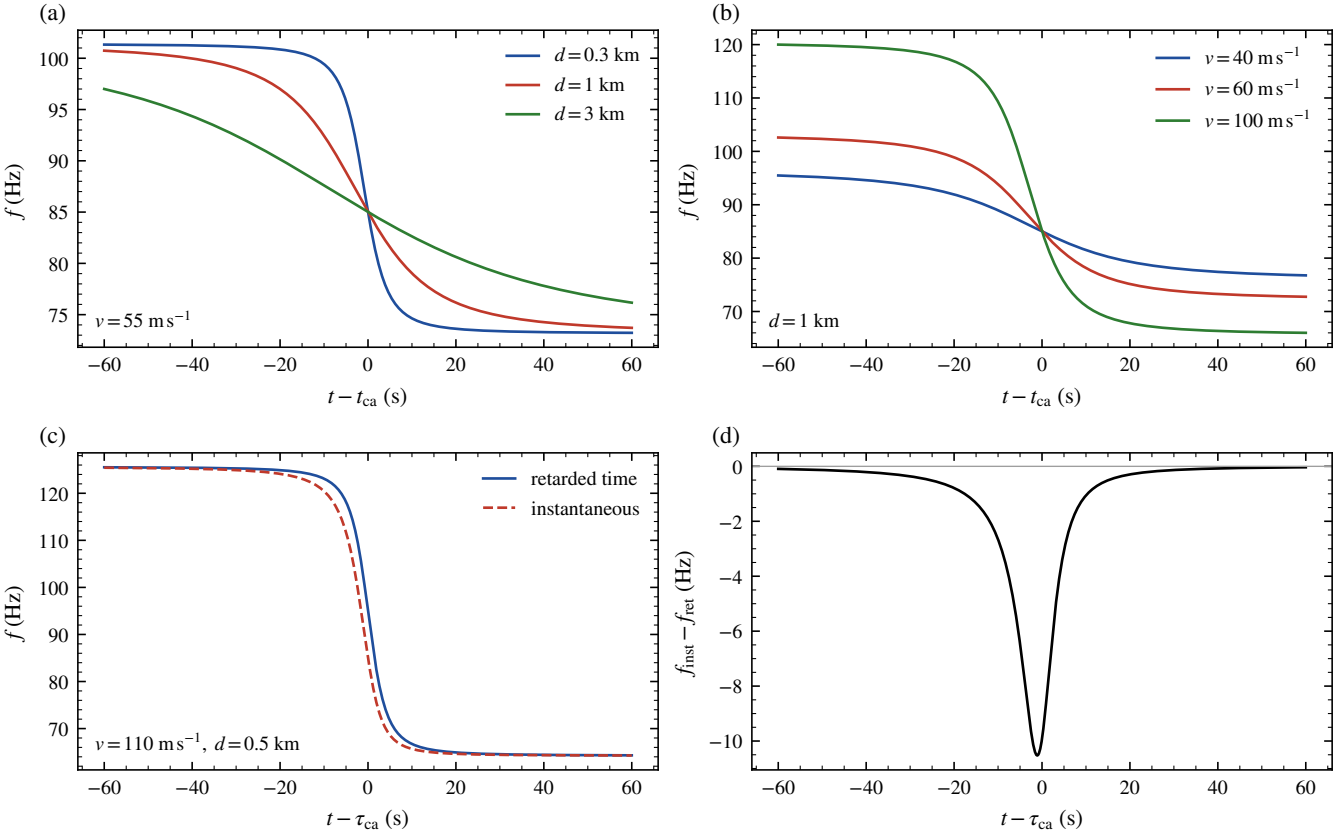
(a)
$d = 0.3$ km
$d = 1$ km
$d = 3$ km
$v = 55$ m s^{-1}
f (Hz)
$t - t_{ca}$ (s)
(b)
$v = 40$ m s^{-1}
$v = 60$ m s^{-1}
$v = 100$ m s^{-1}
$d = 1$ km
f (Hz)
$t - t_{ca}$ (s)
(c)
retarded time
instantaneous
$v = 110$ m s^{-1}, $d = 0.5$ km
f (Hz)
$t - \tau_{ca}$ (s)
(d)
$f_{inst} - f_{ret}$ (Hz)
$t - \tau_{ca}$ (s)